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6.9 Answer Explanations: Two-Part Analysis

The following discussion of Data Insights is intended to familiarize you with the most efficient and effective approaches to the kinds of problems common to Data Insights. The particular questions in this chapter are generally representative of the kinds of Data Insights questions you will encounter on the GMAT exam. Remember that it is the problem-solving strategy that is important, not the specific details of a particular question.

Questions 410 to 427 — Difficulty: Easy

410. Trains M and N are traveling west on parallel tracks. At exactly noon, the front of Train M, which is traveling at a constant speed of 80 kilometers per hour (km/h), is at the rail crossing at Location X, and the front of Train N, which is traveling at a constant speed of 65 km/h, is 30 km west of the rail crossing at Location X. The trains continue traveling at their respective speeds until the front of Train M and the front of Train N are simultaneously at the rail crossing at Location Y.

    In the table, identify the number of kilometers that the front of Train M has traveled between noon and 12:45 p.m. and the number of kilometers that the front of Train N has traveled between noon and 1:00 p.m.

Front of Train M	Front of Train N	
		55
		60
		65
		70
		75

Answer Explanation:

Infer

***Front of Train M*:**

The elapsed time between noon and 12:45 p.m. is 45 minutes, which is $\frac{3}{4}$ hour. In traveling at 80 kilometers per hour for $\frac{3}{4}$ hour, the front of Train M traveled $\frac{3}{4}\left(80\right) = 60$ kilometers.

The correct answer is *60*.

***Front of Train N*:**

The elapsed time between noon and 1:00 p.m. is 1 hour. In traveling at 65 kilometers per hour for 1 hour, the front of Train N traveled 1(65) = 65 kilometers.

The correct answer is *65*.

411. An international basketball tournament will be held in either Nation QN or Nation RN. Exactly six nations, including the host, plan to participate, depending on the following conditions:

    SN will participate only if TN does. UN will not participate if either VN or WN does. WN will not participate unless the tournament is held in RN.

    Based on the information provided, and assuming WN participates, in the first column select the nation that must also participate, and in the second column select the nation that will not participate. Make only two selections, one in each column.

Must participate	Will not participate	
		QN

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		QN
		RN
		SN
		TN
		UN
		VN

Answer Explanation:

Apply

***Must participate*:**

We are told to assume that WN participates. If so, we know that the tournament will be held in RN, because WN will not participate unless that is where the tournament is held. And if the tournament is held in RN, then it must be that RN will participate, because the host nation will participate in the tournament. Therefore, assuming WN participates, it must be the case that RN will participate as well.

The correct answer is *RN*.

***Will not participate*:**

We are told to assume that WN participates. We are also told that UN will not participate if either VN or WN does. So, assuming that WN participates, it cannot be that UN participates.

The correct answer is *UN*.

412. A portion of an automobile test track is divided into Segment A, Segment B, and Segment C, in that order. In a performance test on a car, the car traveled Segment A at a constant speed of 140 kilometers per hour (km/h). Immediately after this, the car rapidly slowed on Segment B and then traveled on Segment C at a constant speed of 70 km/h. The length of Segment C is 3 times the length of Segment A, and it took a total of 42 minutes for the car to travel both Segments A and C. In the table, select the length of Segment A, in kilometers, and select the length of Segment C, in kilometers. Make only two selections, one in each column.

Length of Segment A (kilometers)	Length of Segment C (kilometers)	
		8
		14
		24
		42
		72
		126

Answer Explanation:

Strategize

Length of Segment A (kilometers)

We are asked to determine the lengths of two segments of an automobile test track. The passage states that a car traveled Segment A at a constant rate of 140 km/h and traveled Segment C at a constant rate of 70 km/h, which indicates that it took the car twice as much time to cover a given unit of distance in Segment C as it took to cover the same distance in Segment A. The passage further states that the length of Segment C is 3 times the length of Segment A. Because the car took twice the time to cover a given distance in Segment C as in Segment A, and because Segment C is 3 times as long as Segment A, it can be further determined that the car spent 6 times as much time traveling Segment C as it did traveling Segment A. The passage indicates that it took a total of 42 minutes for the car to travel both Segments A and C, and 42 minutes is equal to 0.7 h (42/60). Therefore, given that the car spent 6 times as much of the total combined time on Segment C as it did on Segment A, it can be calculated that the car spent 0.1 h on Segment A and 0.6 h on Segment C. Because we know that the car was traveling at 140 km/h for 0.1 h on Segment A, we can determine that Segment A was 14 kilometers long.

The correct answer is *14*.

Length of Segment C (kilometers)

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Length of Segment C (Kilometers)

As noted in the analysis above, the car spent 0.6 h traveling Segment C. Because we know that the car was traveling at 70 km/h for 0.6 h on Segment C, we can determine that Segment C was 42 kilometers long.

The correct answer is *42*.

413. Let X, Y, and Z denote the number of international students, in thousands, that Company U predicted would be studying in the United States (US) during the school years 2014–2015, 2019–2020, and 2024–2025, respectively. The average (arithmetic mean) of X, Y, and Z is 1,128, and $Y = 1{,}124$. Company U predicted that there would be more international students studying in the US during the 2024–2025 school year than during the 2014–2015 school year.

In the table, identify a value of X and a value of Z that are jointly consistent with the information provided. Make only two selections, one in each column.

X	***Z***	
		910
		995
		1,175
		1,350
		1,435

Answer Explanation:

Apply

Infer

The task is to determine values for X and Z, where $X < Z$, such that the arithmetic mean of X, Y, and Z is 1,128 and $Y = 1{,}124$. Given that the arithmetic mean of X, Y, and Z is 1,128, it follows that $\frac{X+Y+Z}{3}$ and $X + Y + Z = 3(1{,}128) = 3{,}384$. If $Y = 1,\ 124$, then $X + Z = 3,\ 384 - 1,\ 124 = 2,\ 260$. Since $X < Z$ and $\frac{2,\ 260}{2}$, it follows that $X < 1,\ 130$ and $Z > 1,\ 130$.

***X*:**

Since $X < 1{,}130$, the only table values possible are $X = 910$ or $X = 995$. If $X = 910$, then, since $X + Z = 2{,}260$, $Z = 2,\ 260 - 910 = 1,\ 350$, which is a table entry.

The correct answer is *910*.

***Z*:**

Since $X = 910$ and $X + Z = 2,\ 260$, it follows that $Z = 2,\ 260 - 910 = 1,\ 350$.

The correct answer is *1,350*.

414. Advice from a computer security expert, on passwords used for accessing online accounts follows:

Computer users should use a different password for each online account. They should also use *strong passwords*, which are hard for hackers to decipher. However, strong passwords are difficult to remember, and this is especially the case for users who have multiple online accounts.

Fortunately, software is available at no cost that can store and encrypt a user's passwords: the user need only remember one password to access the rest. For users willing to take the time to install the software on a computer and enter all the required data, such software provides one way to comply with my advice. Some versions of the software can even be copied to a portable device, such as a USB drive, whereby a user can access passwords from any compatible computer.

For computer users with multiple accounts who would not otherwise use a different strong password on every account, the expert's advice amounts to suggesting that such users make certain sacrifices in order to make certain gains. Indicate by appropriate selections in the first and second column which of the items in the third column such users would sacrifice and which they would gain. Make only two selections, one in each column.

Sacrifice	**Gain**	

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Sacrifice	Gain	
		Money
		Computing speed
		Security
		Number of online accounts
		Time

Answer Explanation:

Infer

***Sacrifice*:**

To be able to follow the computer security expert's advice effectively, the expert acknowledges that computer users with multiple accounts will have to make a sacrifice of time to install the software and input the relevant data.

The correct answer is *Time*.

***Gain*:**

The computer security expert claims that making the sacrifice of time involved in installing the software and inputting the relevant data will result in an increase in security (namely, by making online accounts secure from hacking).

The correct answer is *Security*.

415. Some studies of how it was that Prussia resisted the wave of European democratization in the nineteenth century point to Prussia's unequal distribution of land ownership as the principal explanation. And political figures from areas in Prussia with high inequality in land ownership tended to be the staunchest opponents of democratic reforms.

However, **arguments that focus only on those figures' conflicts with landless people ignore the fact that political figures are not only representatives of socioeconomic interests but also political actors embedded in particular contexts that shape whether they support democratic reforms**. Historians must take such contexts into account if they hope to explain what transpired in Prussia during this period.

Select *Passage* for the statement that best describes the purpose of the passage as a whole, and select *Portion in **Boldface*** for the statement that best describes the purpose of the portion in **boldface**. Make only two selections, one in each column.

Passage	Portion in Boldface	
		To offer an alternative to the standard explanation of a particular historical event
		To provide evidence that economic issues were at the root of a particular historical event
		To criticize a certain type of argument about a particular historical event
		To criticize the motives of political figures involved in a particular historical event
		To encourage historians not to approach a particular historical event from too narrow a perspective

Answer Explanation:

Evaluate

The passage asserts that some studies appeal to the fact that Prussia had significant inequality in land ownership during the nineteenth century to explain why Prussia did not participate in the wave of democratization in that century. Political figures in areas of Prussia with high inequality of land ownership were often most strongly opposed to democratization. The passage, however, asserts (in the **boldfaced** text) that such arguments ignore that these political figures not only represent socioeconomic interests but also are political actors embedded in contexts that affect whether they support democratization. The passage claims that historians must take such contexts into account to arrive at an accurate understanding of Prussia during this period.

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***Passage*:**

The passage argues that historians should not focus too narrowly on socioeconomic interests. Instead, the passage asserts that they should think more broadly and consider other contexts.

The correct answer is *To encourage historians not to approach a particular historical event from too narrow a perspective.*

***Portion in Boldface*:**

The portion in **boldface** is critical of arguments that focus only on political figures' conflicts with landless people, saying that such arguments ignore certain key facts about the political figures.

The correct answer is *To criticize a certain type of argument about a particular historical event.*

416. In September 2010, Armando purchased 5 identically priced shirts and 3 identically priced pairs of trousers from a certain department store. Armando paid a total of €28.85 for the shirts and €38.97 for the trousers. In October 2010, the department store held a sale, taking 25% off the September price of everything in the store. Armando returned to the store during the October 2010 sale and bought 11 of the shirts and 7 pairs of the trousers for some friends.

In the table, select how much Armando paid in October for the 11 shirts and 7 pairs of trousers, rounding for each item to the nearest cent. Make only one selection in each column.

Shirts	Trousers	
		€47.60
		€63.51
		€68.20
		€79.39
		€90.91
		€204.54

Answer Explanation:

Infer

***Shirts*:**

In September 2010, Armando purchased 5 identically priced shirts for a total of €28.85. It follows that he paid $€\frac{28.85}{5} = €5.77$ each. At the sale in October 2010, he purchased 11 of these shirts at a reduced price of 25% off the September 2010 price and paid a total of (€5.77)(0.75)(11) ≈ €47.60.

The correct answer is *€47.60*.

***Trousers*:**

In September 2010, Armando purchased 3 identically priced pairs of trousers for a total of €38.97. It follows that he paid $€\frac{38.97}{3} = €12.99$ each. At the sale in October 2010, he purchased 7 of these pairs of trousers at a reduced price of 25% off the September 2010 price and paid a total of (€12.99)(0.75)(7) ≈ €68.20.

The correct answer is *€68.20*.

417. A city is hosting a table tennis tournament for its residents. Each team has exactly two players, and each player is on exactly one team. In each round, each team plays exactly one other team and either wins or loses. The winning team advances to the next round and the losing team is eliminated. No team or player drops out except by losing a game. The tournament is in progress, and exactly 512 players participated in the first round.

From the available options, select a number of tournament rounds and a number of teams such that after the specified number of rounds there will be the specified number of teams remaining in the tournament. Make only two selections, one in each column.

Rounds Completed	Teams Remaining	
		2
		4

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		4
		8
		16
		32

Answer Explanation:

Apply

Before any rounds are completed, there are 512 players, each on exactly one team of two players. Therefore, there are 256 teams in the competition. After each round, half of the teams are eliminated. The following table summarizes the results after each round of the tournament.

Rounds completed	1	2	3	4	5	6	7	8
Teams remaining	128	64	32	16	8	4	2	1

The only available combination of rounds completed and number of teams remaining is 4 rounds completed and 16 teams remaining.

***Rounds completed*:**

The correct answer is *4*.

***Teams remaining*:**

The correct answer is *16*.

418. Adiliah, Bao, Davi, Laszlo, Saleema, and Yarah work in a firm's legal department. Adiliah supervises Bao and Davi, Davi supervises Laszlo, and Laszlo supervises Saleema and Yarah. These are the only supervisory relationships involving these six employees. Each document that the department processes must be initially reviewed by exactly one department member. Each document reviewed by a department member must then be reviewed by that person's supervisor. No other rules require anyone else to review any document. Anyone not required to review a given document will not review it.

Select *Laszlo among reviewers* for the maximum number of department members who could have reviewed a single document if Laszlo was among the reviewers. Select *Adiliah among reviewers* for the maximum number of department members who could have reviewed a single document if Adiliah was among the reviewers. Make only two selections, one in each column.

Laszlo among reviewers	**Adiliah among reviewers**	
		1
		2
		3
		4
		5
		6

Answer Explanation:

Apply

***Laszlo among reviewers*:**

If Laszlo reviews a document, so must his supervisor, Davi, and Davi's supervisor, Adiliah. If Laszlo is the initial reviewer, then exactly 3 department members would review the document. However, if Laszlo reviews a document but is not the initial reviewer, the initial reviewer must be either Saleema or Yarah for a maximum total of 4 reviewers of the document.

The correct answer is *4*.

***Adiliah among reviewers*:**

If Adiliah reviews a document, no department members will review the document after her. If Adiliah is the initial reviewer, then exactly 1 department member will review the document. If Bao or Davi is the initial

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reviewer, then the document will be reviewed by exactly 2 department members. If Laszlo is the initial reviewer, then the document will be reviewed by exactly 3 department members. Finally, if Saleema or Yarah is the initial reviewer, then exactly 4 department members will review the document. Since no one else can be the initial reviewer, 4 must be the maximum number of reviewers of a single document for which Adiliah is among the reviewers.

The correct answer is *4*.

419. Naturalist: The decline of coral reefs has various causes. One contributing factor is predation on coral by organisms such as the crown-of-thorns sea star, whose preferred food source is coral polyps. Human fishing practices have decreased the sea star's predators, such as the harlequin shrimp. It is also possible that runoff containing nutrients for phytoplankton has resulted in larger phytoplankton blooms: the crown-of-thorns sea star gladly eats phytoplankton.

Indicate which cause-and-effect sequence would most likely, according to the naturalist, result in coral reef decline. Make only two selections, one in each column.

Cause	Effect	
		An increase in phytoplankton
		A decrease in phytoplankton
		An increase in crown-of-thorns sea stars
		A decrease in crown-of-thorns sea stars
		An increase in harlequin shrimp

Answer Explanation:

Infer

We are asked to provide a sequence of cause and effect that would (according to the naturalist) result in coral reef decline. The question suggests that the answers will be dependent on one another, in that the correct answer for *Cause* will depend on the correct answer for *Effect*, and vice versa.

The passage states that one factor contributing to coral reef decline is predation on coral by organisms such as the crown-of-thorns sea star. The preferred food source for this organism is coral polyps, and so a causal sequence that has as its effect an increase in crown-of-thorns sea stars may be likely to produce a decline in coral reefs. This suggests that the third option may be the correct response for *Effect*. However, this depends on whether one of the other statements in the table describes a cause suggested by the naturalist to produce this effect.

The last sentence of the naturalist's statement indeed suggests, in the context of the passage, that an increase in phytoplankton blooms would result in greater numbers of crown-of-thorns sea stars. This suggests that the first option may be the correct response for *Cause*. And indeed, according to the naturalist, the crown-of-thorns sea star's preferred food source is coral polyps. All of the naturalist's statement is about possible explanations of the decline of coral reefs. The passage therefore indicates that the naturalist would take an increase in phytoplankton (the first option, *Cause*) to produce an increase in crown-of-thorns sea stars (the third option, *Effect*), and that this causal sequence would likely result in coral reef decline.

***Cause*:**

The correct answer is *An increase in phytoplankton*.

***Effect*:**

The correct answer is *An increase in crown-of-thorns sea stars*.

420. Archaeologist: There were several porcelain-production centers in eighteenth-century Britain, among them Bristol, Plymouth, and New Hall. Each center developed a unique recipe for its porcelain that might include flint glass, soapstone, bone ash, clay, quartz, and so on. We will therefore be able to determine, on the basis of compositional analysis, where the next cup we recover from this archaeological site was made.

Indicate two different statements as follows: one statement identifies an *assumption required* by the archaeologist's argument, and the other identifies a *possible fact* that would, if true, provide significant logical support for the required assumption. Make only two selections, one in each column.

Assumption	Possible	

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Assumption Required	Possible Fact	
		Other cups have been recovered from the archeological site, all of which were made of porcelain.
		Some of the cups recovered from the archeological site were not made of porcelain.
		The next cup to be recovered from the site will likely be made of porcelain.
		Porcelain makers often traveled between centers, experimenting with one another's recipes.
		There was considerable overlap of materials in the recipes used by the various centers.
		Most porcelain in eighteenth-century Britain was made at one of the several centers.

Answer Explanation:

Infer

This item asks for a possible fact and an assumption of the argument such that the possible fact supports the assumption. This suggests that the correctness of the answer for one response opportunity could depend on the correctness of the answer for the other.

***Assumption Required*:**

Only one of the statements in the table is an assumption required by the archaeologist's argument. If the next cup recovered from the archaeological site is not made of porcelain, then the evidence provided by the archaeologist has no relevance regarding where the next cup recovered from the site was made. This indicates that the third statement may be the correct response.

In contrast, the first statement, although a strengthener of the argument, is not a required assumption: we could, for example, have other reasons for believing that the next cup will be made of porcelain. The sixth statement would also be an important strengthener, but it is not required for the reasoning of the argument to be sound.

The correct answer is *The next cup to be recovered from the site will likely be made of porcelain*.

***Possible Fact*:**

Given that the third statement is the required assumption within the list in the table, the possible fact asked for in the question must be a possible fact that provides significant logical support for this statement. (Because the question specifies that two *different* statements must be selected, the *possible fact* cannot be the third statement.)

The correct answer is *Other cups have been recovered from the archaeological site, all of which were made of porcelain*.

421. Health advocate: The government's current farm-subsidy system primarily rewards large farms for planting monocultures of corn, soybeans, wheat, and rice. Most of the crops produced in this way go to feed livestock in factory farms, which results in a glut of fatty meats in the marketplace. A large proportion of such crops that are not used to feed livestock are used to make sugary processed foods. These subsidies promote unhealthy diets by making sugary foods and fatty meats artificially cheap. Obviously, it is important for the government to avoid these effects.

On the basis of the information above, select *Recommends* for the option that describes the government action that the health advocate most likely recommends, and select *Intended result* for the option that describes what the health advocate likely hopes will be the result of that action. Make only two selections, one in each column.

Recommends	Intended result	
		Improve the overall quality of livestock feed
		Improve the overall quality of people's diets
		Reduce the overall financial cost of people's diets
		Prevent the manufacture of sugary processed food
		Change the farm-subsidy system

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Answer Explanation:

Infer

***Recommends*:**

The health advocate's comments have little or no relevance to overall **quality** of livestock feed, **overall** financial cost of people's diets, or **prevention of the manufacture** of sugary processed food. Of the two remaining options, the health advocate's comments are more appropriate for an argument to change the farm-subsidy system than to improve the overall quality of people's diets, the latter of which the health advocate's last two sentences suggest would be a consequence of changing the farm-subsidy system.

The correct answer is *Change the farm-subsidy system*.

***Intended result*:**

The explanation above shows that, among the options, improving the overall quality of people's diets is the intended result of changing the farm-subsidy system.

The correct answer is *Improve the overall quality of people's diets*.

422. An archaeologist studying Artifacts A–D is interested in whether Artifact C is older than Artifact A.

Select an assertion involving Artifact A and an assertion involving Artifact C that together imply the assertion "Artifact C is older than Artifact A." Make only two selections, one in each column.

Assertion involving Artifact A	**Assertion involving Artifact C**	
		Artifact A is older than Artifact B.
		Artifact A is older than Artifact D.
		Artifact B is older than Artifact A.
		Artifact B is older than Artifact D.
		Artifact C is older than Artifact D.
		Artifact C is older than Artifact B.

Answer Explanation:

Infer

We are asked to select an assertion pertaining to Artifact A and an assertion pertaining to Artifact C that together imply the assertion *Artifact C is older than Artifact A*.

An assertion that Artifact A is *older* than one of the three other artifacts will not, together with any of the assertions involving Artifact C, imply the assertion *Artifact C is older than Artifact A*. Therefore, the assertion involving Artifact A must be an assertion that one of the other artifacts is older than Artifact A. Only the third option, *Artifact B is older than Artifact A*, is such an assertion.

Given the assertion that *Artifact B is older than Artifact A*, the assertion involving Artifact C that *Artifact C is older than Artifact B* implies that *Artifact C is older than Artifact A*. None of the other pairs of options would together imply that *Artifact C is older than Artifact A*.

***Assertion involving Artifact A*:**

The correct answer is *Artifact B is older than Artifact A*.

***Assertion involving Artifact C*:**

The correct answer is *Artifact C is older than Artifact B*.

423. For a randomly selected day, the probability that a visitor to a certain pond will see at least one swan is 0.35. The probability that a visitor to that pond on a randomly selected day will see at least one heron is 0.2. Furthermore, seeing a swan and seeing a heron are independent of each other.

Based on the information provided, select *Both swan and heron* for the probability that a visitor to the pond will see both at least one swan and at least one heron on any given day, and select *Neither swan nor heron* for the probability that a visitor to the pond will see neither a swan nor a heron on any given day. Make only two selections, one in each column.

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Both swan and heron	Neither swan nor heron	
		0.02
		0.07
		0.48
		0.52
		0.70

Answer Explanation:

Apply

***Both swan and heron*:**

We're told that seeing a swan and seeing a heron are independent of each other. The probability that two independent events both occur equals the probability of the first multiplied by the probability of the second. So, the probability that a visitor to the pond will see both at least one swan and at least one heron on any given day is $0.35 \times 0.2 = 0.07$.

The correct answer is *0.07*.

***Neither swan nor heron*:**

If the probability of an event is P, the probability that the event won't occur is $1 - P$. So, the probability of a visitor to the pond not seeing a swan on any given day is $1 - 0.35 = 0.65$, and the probability of the visitor not seeing a heron on any given day is $1 - 0.2 = 0.8$. Since seeing a swan is independent of seeing a heron, not seeing a swan is likewise independent of not seeing a heron. Thus, the probability that a visitor to the pond will see neither a swan nor a heron on any given day is $0.65 \times 0.8 = 0.52$.

The correct answer is *0.52*.

424. Two primary school teachers are debating the merits of classrooms with highly decorated walls.

Teacher 1: A recent study has shown that highly decorated walls can interfere with learning in the classroom. We should therefore decrease the number of wall decorations in our classrooms.

Teacher 2: The results of that study are correct, but pertain only to the retention of information from single lessons in which the students were in the classroom for only a short period of time. Our students are in the same classroom all day, five days a week. Without some added stimulation, students can become bored and therefore even more distracted than they would be by highly decorated walls. I therefore maintain that highly decorated walls serve to increase learning over the long term.

Select for X and for Y the options such that the following statement, if filled in accordingly, has the most support from the information provided. Make only two selections, one in each column. Teacher 2 suggests that he agrees with Teacher 1 about the effects of wall decorations on ____X____, but disagrees about the effects of wall decorations on ____Y____.

X	Y	
		retention of certain specified types of information
		the effectiveness of teaching methods that eschew rote learning
		the ability of students to maintain skills if practiced in various classrooms
		retention of information from multiple lessons given in the same room over a period of time
		retention of information from lessons given in rooms occupied for only a short time

Answer Explanation:

Infer

***X*:**

Teacher 1 cites a study as having shown that highly decorated walls interfere with classroom learning. Teacher 2 remarks that these study results are correct with regard to retention of information from single, brief lessons. Thus, the two teachers agree that wall decorations interfere with retention of information from lessons given in rooms occupied for only a short time.

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The correct answer is *retention of information from lessons given in rooms occupied for only a short time.*

***Y*:**

By citing the study without qualification as a reason to remove classroom wall decorations, Teacher 1 implicitly accepts that the study results hold in general. Thus, Teacher 1 seems to believe that highly decorated walls generally interfere with classroom learning. But Teacher 2 argues that when students are in the same room for many lessons over long periods, decorated walls reduce their boredom and distraction, helping them learn better in the long run. So, the two teachers evidently disagree about whether highly decorated walls interfere with students' retention of information from multiple lessons given in the same room over time.

The correct answer is *retention of information from multiple lessons given in the same room over a period of time.*

425. Natalya put 14 blue marbles, 14 red marbles, and no other marbles into 3 empty cups. She put 2 blue marbles and 6 red marbles in Cup A, 6 blue marbles and 4 red marbles in Cup B, and 6 blue marbles and 4 red marbles in Cup C. After this, Dmitry will randomly pick three marbles, one from each cup.

Consistent with the given information, select for *3 blue* the probability that Dmitry will pick 3 blue marbles, and select for *3 red* the probability that Dmitry will pick 3 red marbles. Make only two selections, one in each column.

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